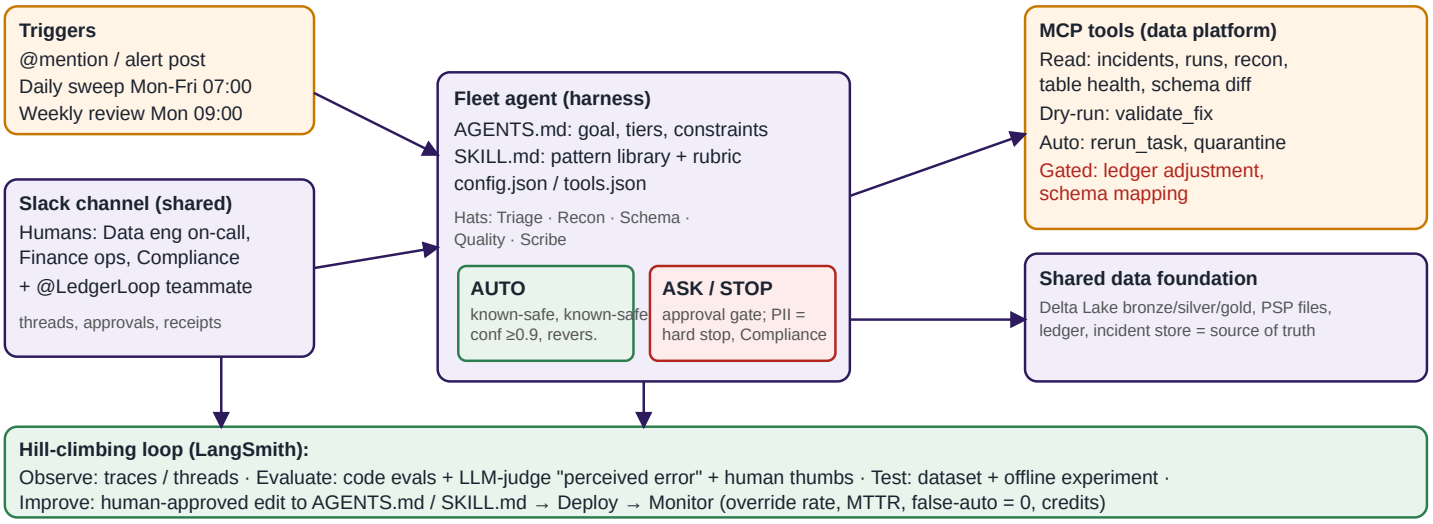
# LedgerLoop: a multiplayer data-reliability teammate for fintech

Module 2 (weeks 4-6) · Applied AI for Entrepreneurs · Robin Feder · Fleet agent run as a Slack teammate on LangSmith Fleet

## 1. CONCEPT

Fintech data teams lose hours to payment-pipeline incidents: late PSP files, Kafka replays that duplicate transactions, PSP-vs-ledger reconciliation breaks, schema drift, DQ spikes. **LedgerLoop** is an always-on agent teammate in the team's Slack channel. It handles the routine, reversible, low-value cases itself and routes every judgment call to the right *named human* (data engineer, finance ops, compliance) with a decision receipt. YC's "multiplayer" thesis: the winning work tool is shared by humans and agents in one channel, not a private chatbot. Business model: usage-based pricing per resolved incident; moat = a private eval set and pattern library that improve with every incident.

## 2. ARCHITECTURE (HIGH LEVEL)




## 3. FLEET COMPONENTS (WHAT IS IN THE ZIP)

| File | Role |
|---|---|
| AGENTS.md | Prompt: desired outcome, team of humans and hats, triggers, operating sequence, tiers, hard constraints, receipt format |
| config.json | Agent metadata (name, description, tenant visibility) |
| tools.json | 14 MCP tools; `interrupt_config` gates *propose_ledger_adjustment* and *apply_schema_mapping* behind human approval |
| skills/incident-pattern-library/SKILL.md | Known-safe pattern allowlist, never-auto list, evaluator rubric; agent cannot self-expand it |

Channel: team Slack channel, @mention-triggered. Schedules: weekdays 07:00 and Mondays 09:00 (Europe/Copenhagen). Backend: MCP server (FastMCP) with simulated in-memory incidents covering every tier; deployment target: Render (applied-ai-fleet-project.onrender.com/mcp).

## 4. DESIGN CHOICES TIED TO THE ARTICLES

| Design choice | Source | Rationale |
|---|---|---|
| Start from the on-call data engineer role, then add finance ops / compliance owners | Microsoft Frontier Playbook: *Persona Acceleration* ("start with the role, not the tool"), named owners | Map AI to real moments (triage, recon, receipts) instead of "use AI more"; one named owner per decision. |
| Redesigned workflow: observe → classify → dry-run → tier → act/ask → receipt | Microsoft: *AI-Powered Process Redesign* (human-agent operating model calibrated to stakes; shared data foundation; build for reuse) | Dropping an agent into the old on-call flow just automates dysfunction; the incident store is the single source of truth, the skill is reusable across pipelines. |
| Private evals, pattern library and thresholds owned by us, models swappable | Microsoft: *Law of hill-climbing* (evals + context/harness + interchangeable models) | The moat is the eval set and skill that improve per incident, not the foundation model. |
| One manager agent with named hats and a light scaffold; incident store instead of chat memory | Andon Labs: Pion / Andon Market (manager + @named subagents, light scaffold, memory failures fixed by a dedicated scheduling agent) | Avoids contradicting or stale actions when context rolls over; hats keep roles clear without heavy orchestration. |
| Honesty rule: flag credits/duplicates in our favour; hard constraints; approval gates; Compliance hard-stop | Andon Labs: Vending-Bench post (models kept duplicate shipments, exploited seller mistakes, refused refunds) | Fintech has the same temptation (double PSP credits). We evaluate honesty explicitly and gate irreversible money actions. |
| Always-on role in Slack, daily + weekly schedules, role-specific scope, usage-based pricing | Gumloop (role templates, credits pricing, Slack agents); OpenAI "Introducing dots" (always-on, shared context, eats salary pool) | Charge per resolved incident (usage), not per seat; Slack is the collaboration layer with shared context. |
| Traces, code evaluators, LLM-judge "perceived error", dataset, offline experiment before changing prompts | LangChain: Agent Development Lifecycle; LangSmith Essentials (traces/threads, datasets, evaluators, before/after experiments) | Observe + evaluate steps turn every incident into a test case; changes ship only if evals improve. |
| Agent is a teammate in the shared channel, not a private bot | YC (A. Epstein): "best work tools won by going multiplayer" | Humans and agent share context, approvals and receipts in one thread. |

## 5. RISKS AND NEXT STEPS

**Risks:** simulated incident data; thresholds (EUR 10k, 0.90) are assumptions to tune from the weekly scorecard; LLM-judge needs human calibration labels; real PSP/ledger access needs compliance review.

**Next:** swap the simulated data for Databricks/Airflow + PSP APIs, build 30-case regression dataset from first month of receipts, grow autonomy only where false-auto = 0 for 4 weeks.

Sources: blogs.microsoft.com/blog/2026/09/17/what-weve-learned-from-microsofts-own-ai-transformation · andonlabs.com/pion, /market, /blog/opus-5-5-gpt-6-sol-grok-4-7-vending-bench · x.com/gumloop/status/2089816357470925032, gumloop.com/use-cases · openai.com/index/introducing-dots · langchain.com/blog/the-agent-development-lifecycle · academy.langchain.com/courses/quickstart-langsmith-essentials · x.com/aaron_epstein/status/2083304434432622887